



Lesson 2.8 — Systems with Quadratics

Big idea: Solving a system with at least one quadratic comes down to substitution. Set the equations equal, get one quadratic, solve, then back-substitute.

Quick review

Linear-quadratic: substitute the linear into the quadratic. Up to 2 solutions (the line crosses the parabola at 0, 1, or 2 points).

Quadratic-quadratic: set them equal to each other. Up to 4 solutions in principle, often fewer.

Always: back-substitute each x into either equation to get y . Don't forget the y -values.

Geometric meaning: solutions are the intersection points.

Worked example

Worked example. Solve the system $y = x^2 - 1$ and $y = x + 1$.

Set equal: $x^2 - 1 = x + 1 \Rightarrow x^2 - x - 2 = 0 \Rightarrow (x - 2)(x + 1) = 0 \Rightarrow x = 2$ or $x = -1$.
 y at $x = 2$: 3. y at $x = -1$: 0. Solutions: (2, 3) and (-1, 0).

Warm-ups

1. Solve: $y = x^2$ and $y = 4$.
2. Solve: $y = x + 2$ and $y = x^2$.
3. Does the line $y = x - 5$ intersect $y = x^2$? (Find solutions or argue why not.)

Practice

1. Solve: $y = x^2 - 4$ and $y = 2x - 1$.
2. Solve: $y = x^2 + 3x - 4$ and $y = 2x - 4$.
3. Solve: $y = x^2$ and $y = -x^2 + 8$.

Challenge

1. For what values of m does the line $y = mx + 3$ intersect $y = x^2 + 4$ at exactly one point? (Hint: discriminant.)
2. A circle $x^2 + y^2 = 25$ and a parabola $y = x^2 - 5$. Find their intersection points.



Answer key — Lesson 2.8

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $x^2 = 4 \Rightarrow x = \pm 2$. Solutions: (2, 4) and (-2, 4).
2. $x^2 = x + 2 \Rightarrow x^2 - x - 2 = 0 \Rightarrow x = 2, -1$. (2, 4) and (-1, 1).
3. $x^2 = x - 5 \Rightarrow x^2 - x + 5 = 0$; $D = 1 - 20 = -19 < 0 \Rightarrow$ no real intersections.

Practice

1. $x^2 - 4 = 2x - 1 \Rightarrow x^2 - 2x - 3 = 0 \Rightarrow (x - 3)(x + 1) = 0$. (3, 5) and (-1, -3).
2. $x^2 + 3x - 4 = 2x - 4 \Rightarrow x^2 + x = 0 \Rightarrow x(x + 1) = 0$. (0, -4) and (-1, -6).
3. $x^2 = -x^2 + 8 \Rightarrow 2x^2 = 8 \Rightarrow x = \pm 2$. (2, 4) and (-2, 4).

Challenge

1. $mx + 3 = x^2 + 4 \Rightarrow x^2 - mx + 1 = 0$. $D = m^2 - 4 = 0 \Rightarrow m = \pm 2$.
2. Substitute $y = x^2 - 5$ into circle: $x^2 + (x^2 - 5)^2 = 25$. Let $u = x^2$: $u + u^2 - 10u + 25 = 25 \Rightarrow u^2 - 9u = 0 \Rightarrow u(u - 9) = 0 \Rightarrow u = 0$ or 9. So $x = 0 (\rightarrow y = -5)$ or $x = \pm 3 (\rightarrow y = 4)$. Points: (0, -5), (3, 4), (-3, 4).